

**Maximum
Temperature Data**



**Identifying
Extreme
Heatwaves**



**Characterizing
Space-
Time Typologies**



**Process-based
Compound
Heatwaves**

Obtain daily maximum temperature (Tmax) for the warm season (Apr–Sep: 1940–2025) over the Eastern Mediterranean & Middle East (EMME)



Identify P95th Tmax exceedances, followed by major event selection for a coverage greater than or equal to a regional P97.5th threshold



Select heatwave events based on their spatial connectivity and clustering into compound heatwaves



Analyze compound heatwave properties: centroid, areal extent, azimuth orientation, axis ratio (L2/L1), duration, frequency